Collision energy revisited

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Learning target: Use energy as well as momentum to classify a collision.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 14.*
15 N toward the natural length.

02. Worked example: explain the model
Total momentum 2 both times; K 2 to 1 J, remainder internal/thermal/sound.

03. Read and interpret the model
Total momentum zero both before after; kinetic energy decreases to zero after sticking.

04. Apply the physics
Each 2 J; total 4 J; final 0 J.

05. Explain your reasoning
Sticking is only perfectly inelastic; bouncing can still transform kinetic energy.

06. Review: connect an earlier idea *Review: Unit 4, Day 5.*
32 J, using $K_f=K_i+W_{net}$.

07. Exit check
Inelastic; thermal, deformation, sound, etc.

08. STRETCH: test the model
Total momentum = 0, so final velocity = 0. Initial K = 4.5 + 4.5 = 9 J, final K = 0. Compared with question 4’s 4 J, 9/4 as much kinetic energy becomes internal energy or other transfers.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 14.*
F=-kx=+7.5 N, right.

02. Worked example: explain the model
Total momentum 2 both times; K 2 to 1 J, remainder internal/thermal/sound.

03. Read and interpret the model
Total momentum zero both before after; kinetic energy decreases to zero after sticking.

04. Apply the physics
vf 2 m/s; 9 J; 6 J; 3 J transformed.

05. Explain your reasoning
Sticking is only perfectly inelastic; bouncing can still transform kinetic energy.

06. Review: connect an earlier idea *Review: Unit 4, Day 5.*
Net work 72 J; v_f=6 m/s from $\frac{1}{2}mv^2=W_{net}$.

07. Exit check
Momentum 4 both; K before 8 J, after 5 J; inelastic.

08. STRETCH: test the model
v_f=2 m/s. K_i=18 J; K_f=6 J; loss 12 J or 2/3.